SARA BAKHTIARI

PhD Candidate in Mechanical Engineering | Scientific Machine Learning | Computational Modeling | Multimodal AI

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Eligible for U.S. internships through F-1 Curricular Practical Training (CPT), subject to university authorization.

PROFESSIONAL SUMMARY

PhD candidate in Mechanical Engineering at the University of Kentucky working across scientific computing, computational biomechanics, and machine learning. Develops physics-based simulations and data-driven workflows using vision-language-model embeddings, learned bottlenecks, regression, repeatability analysis, and cross-validation, with experience in Python, HPC, numerical modeling, and large scientific datasets.

EDUCATION

PhD, Mechanical Engineering (Direct PhD) - University of Kentucky | Spring 2024-Present
Candidate since Apr 2026 | GPA 3.91/4.00 | Advisor: Jonathan F. Wenk, PhD
B.Sc., Mechanical Engineering - Iran University of Science and Technology (IUST) | 2016-2021

RESEARCH EXPERIENCE

Graduate Research Assistant - Computational Cardiac Mechanics | Spring 2024-Present
University of Kentucky - Lexington, KY

- Develop large-scale finite-element cardiac simulations in Python/FEniCS that couple mechanics, active contraction, circulation, and spatial biological state variables.
- Design reproducible HPC workflows using MPI, PETSc, SLURM, structured simulation outputs, and automated post-processing for high-dimensional spatial/time-series data.
- Develop computational studies of constitutive uncertainty, fiber disarray, mechanical efficiency, and physiological compensation across local and organ scales.
- Couple an agent-based fibrosis model with finite-element mechanics, creating a multiscale framework linking spatial biological evolution to tissue mechanics.

Machine Learning & Vision-Language Models for Electrospinning | 2026

- Apply pretrained and fine-tuned vision-language models to electrospinning videos and extract high-dimensional hidden-state embeddings for downstream prediction.
- Develop learned low-dimensional bottleneck representations and compare them with full embeddings and randomly selected features.
- Design repeated embedding-generation, feature-selection, and cross-validation studies to quantify representation variability and predictive robustness.
- Build regression pipelines for predicting electrospun fiber diameter from video-derived representations using Python and scikit-learn.

SELECTED RESEARCH PROJECTS

Physics-Based Cardiac Simulation: Develop nonlinear finite-element models coupling mechanics, active contraction, circulation, and spatial biological variables.

Multiscale Fibrosis Modeling: Link agent-based spatial fibrosis evolution to finite-element tissue mechanics within a coupled computational framework.

Scientific Data Pipelines: Automate post-processing and analysis of high-dimensional spatial and time-series simulation outputs using Python and HPC workflows.

PUBLICATIONS & RESEARCH OUTPUTS

Islam, R., Bakhtiari, S., Anugrah, G. G., Mehri, M., & Wenk, J. F. "Advancements in finite element modeling of myocardial infarction: a review." Biomechanics and Modeling in Mechanobiology, 25(4), 2026. DOI: 10.1007/s10237-026-02107-0

Bakhtiari, S., Mehri, M., & Wenk, J. F. "Choice of Passive Myocardial Constitutive Law Drives Divergent Predictions of Left Ventricular Remodeling in Volume-Overload Conditions." Manuscript under review.

Bakhtiari, S., Mehri, M., & Wenk, J. F. "Effect of Myocardial Fiber Disarray on Left Ventricular Mechanical Efficiency and Baroreflex Compensation." Manuscript under review.

Selected presentation: "Choice of Passive Myocardial Constitutive Law Drives Differences in LV Mechanics and Growth Under Disease Conditions." Poster, BMES Annual Meeting, San Diego, CA, October 2025.

TEACHING EXPERIENCE

Teaching Assistant - Statics, University of Kentucky | Fall 2026

- Solved and explained undergraduate statics problems involving equilibrium equations, free-body diagrams, force/moment analysis, trusses, frames, and systematic engineering problem-solving methods.
- Supported classroom instruction, student questions, engagement, and class management; communicated technical concepts to students with different levels of understanding.
- Assisted with grading and examination supervision, evaluating engineering solution approaches and maintaining consistent exam procedures.

Teaching Assistant - Statics, Iran University of Science and Technology | 2021

- Supported undergraduate engineering-mechanics problem solving and helped students strengthen core statics concepts.

TECHNICAL SKILLS

Machine Learning: scikit-learn, regression, cross-validation, feature selection, embeddings, dimensionality reduction, learned bottlenecks, vision-language models

Programming & Data: Python, NumPy, SciPy, Pandas, MATLAB, structured scientific data processing
Scientific Computing & HPC: FEniCS/DOLFIN, UFL, nonlinear simulation, MPI, mpi4py, PETSc, SLURM, Linux, Singularity
Domain Modeling: Computational biomechanics, cardiovascular mechanics, constitutive modeling, multiscale fibrosis modeling